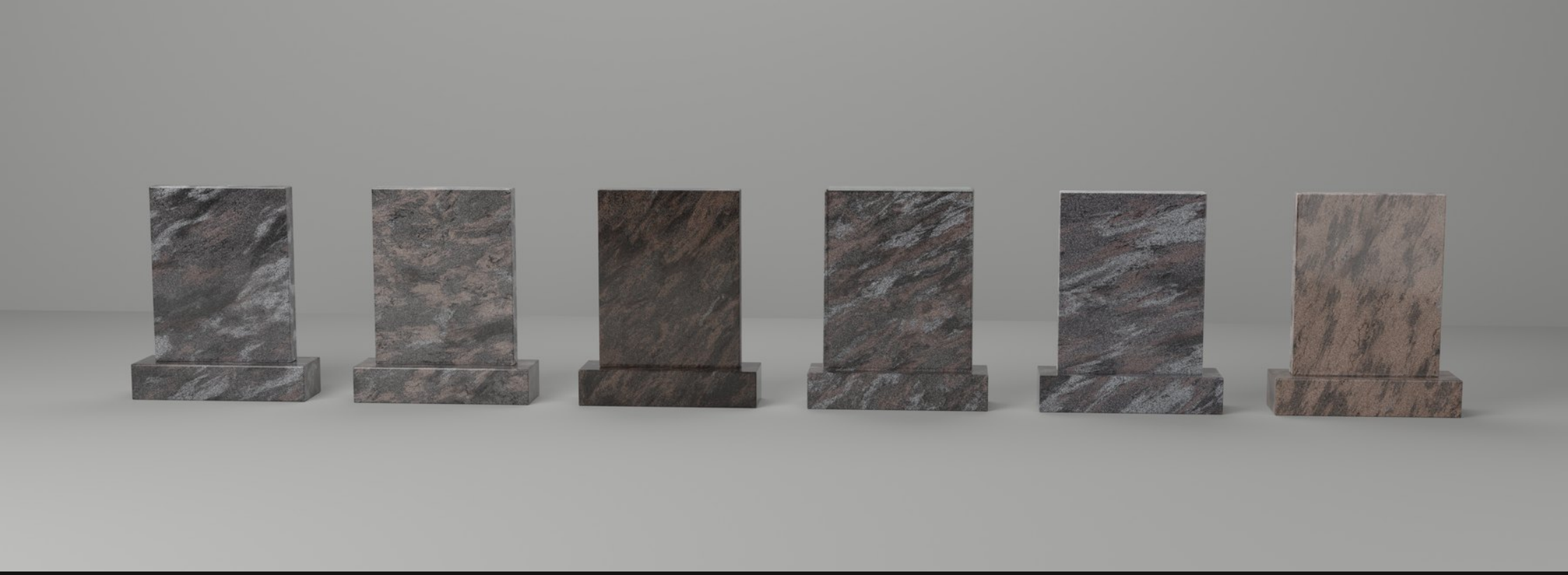


MATERIAL REPORT · SIX DATASETS

Procedural Headstone Granite

Seed-controlled Blender materials derived from photographs of six Paradiso-type gneiss memorials, calibrated against the photos and rendered in HD.

01
Salt & Pepper
folder 1st

02
Flame Swirl
folder 2nd

03
Dark Maroon
folder 3rd

04
Folded + Rock-pitched
folder 4th

05
High Contrast
folder 5th

06
Rust Red
folder 6th

Blender 5.2 LTS · Cycles · October 2026

OVERVIEW

Method and calibration

how the materials were built and checked

What the stone is

All six memorials are cut from the same family of migmatite gneiss sold as "Paradiso" or "Himalayan Blue" granite. Each material is built from the same seven geological layers, with per-dataset presets controlling colour, proportions and flow.

Matrix	fine ~1.5 mm crystals (Voronoi)
Melanosome	broad darker biotite-rich layers
Rose layers	garnet / K-feldspar clouds
Leucosome	grey quartz-feldspar clouds
Speckle	1–4 mm black biotite clots
Schlieren	thin broken black streaks
Lamination	fine foliation stripes along folds

Every layer is sampled in one folded flow space: rotate to the flow angle, stretch, then three levels of domain warping (sweep, fold, crinkle).

How it was checked

For each folder a 23–31 cm patch of clean polished stone was cut from a photo. The shader’s albedo was rendered at the same physical size with an orthographic camera, and the presets were tuned until colour, contrast, percentiles and redness matched.

Calibration: photo vs shader (same patch)

FOLDER	MEAN SRGB	CONTRAST Σ	P5–P95	REDDISH
01 1st	58 55 55	27.6	10–101	10%
photo / shader	**58 54 55**	**23.2**	**13–96**	**9%**
02 2nd	81 76 75	22.5	41–114	18%
photo / shader	**76 71 70**	**23.4**	**31–112**	**24%**
03 3rd	47 41 36	18.3	10–71	52%
photo / shader	**49 41 37**	**17.1**	**13–71**	**51%**
04 4th	59 57 56	25.7	13–98	9%
photo / shader	**55 52 51**	**23.5**	**12–93**	**12%**
05 5th	69 66 68	33.9	11–125	15%
photo / shader	**73 69 72**	**31.6**	**16–129**	**16%**
06 6th	82 69 62	27.1	27–116	94%
photo / shader	**84 71 65**	**24.3**	**31–113**	**93%**

Upper value in each pair is the photo, lower is the shader. Contrast σ is the luminance standard deviation (sRGB 0–255). The first version had σ ≈ 7 and periodic banding; v2 fixes both.

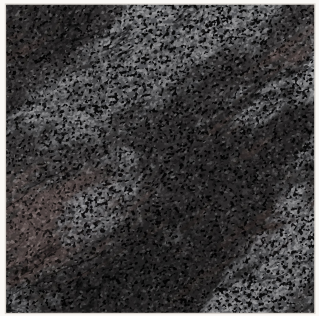
OVERVIEW

Photo vs shader at a glance

each pair covers the same 23–31 cm of stone

01 Salt & Pepper

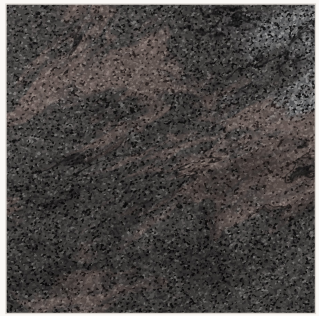
PHOTO



SHADER

02 Flame Swirl

PHOTO



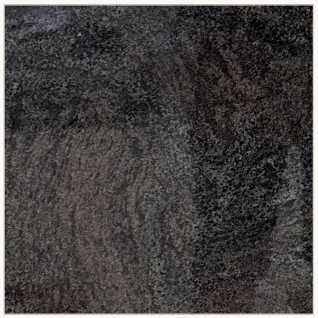
SHADER

03 Dark Maroon

PHOTO

SHADER

04 Folded + Rock-pitched



PHOTO

SHADER

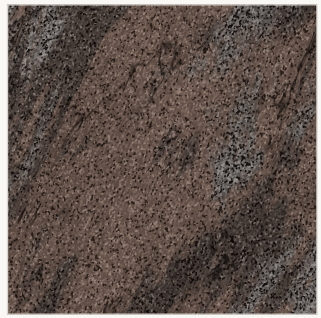
05 High Contrast

PHOTO

SHADER

06 Rust Red

PHOTO



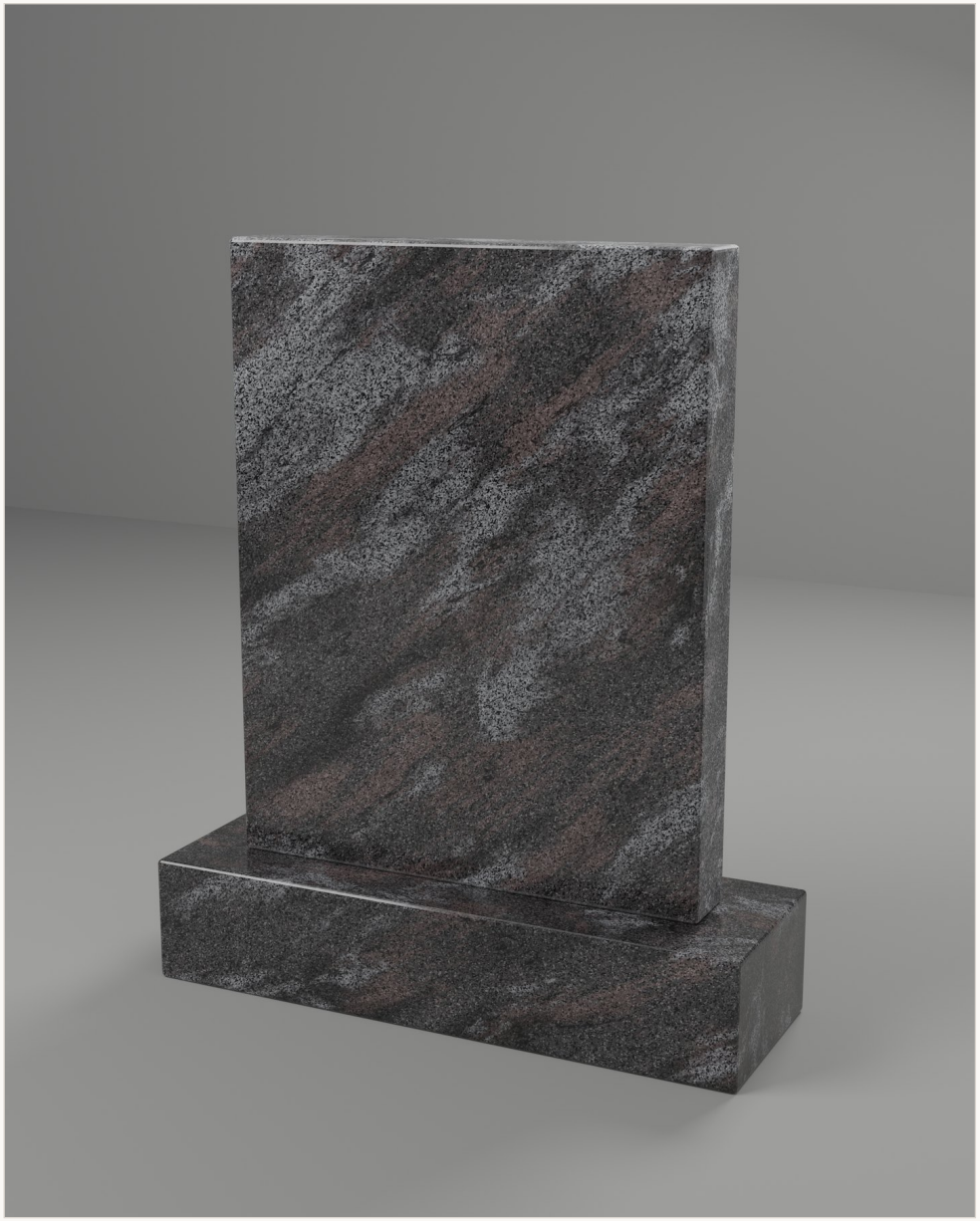
SHADER

Photo patches are cropped from the original 12 MP photographs at the exact calibration area; shader patches are 2048 px albedo renders of the same physical size. The shader has no lighting, reflections or lens blur, so it reads slightly crisper than the photos. Full-resolution versions of every pair appear on each dataset’s surface-detail page.

Salt & Pepper

GS01_SaltPepper · seed 1

REFERENCE PHOTOGRAPH folder 1st · original 12 MP



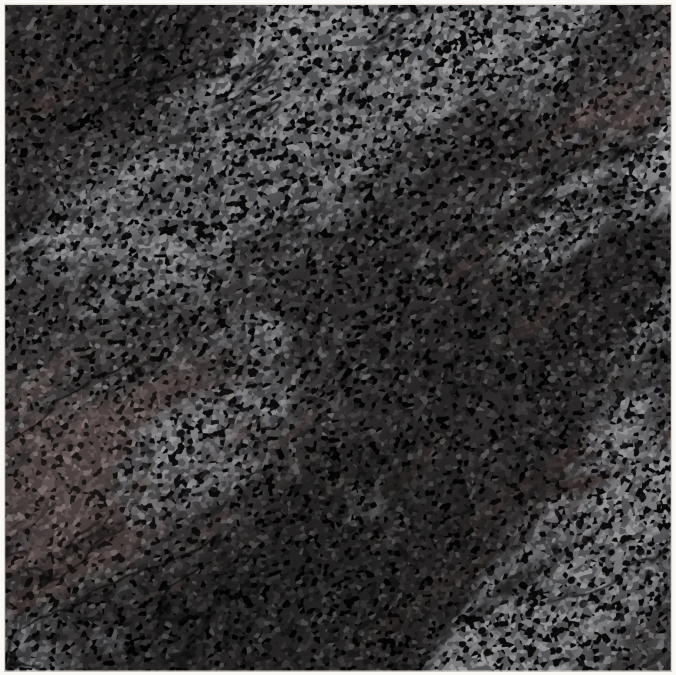
PROCEDURAL MATERIAL Cycles render · 2400 × 3000 · polished + honed

DATASET 01 · FOLDER 1st · REF 269567

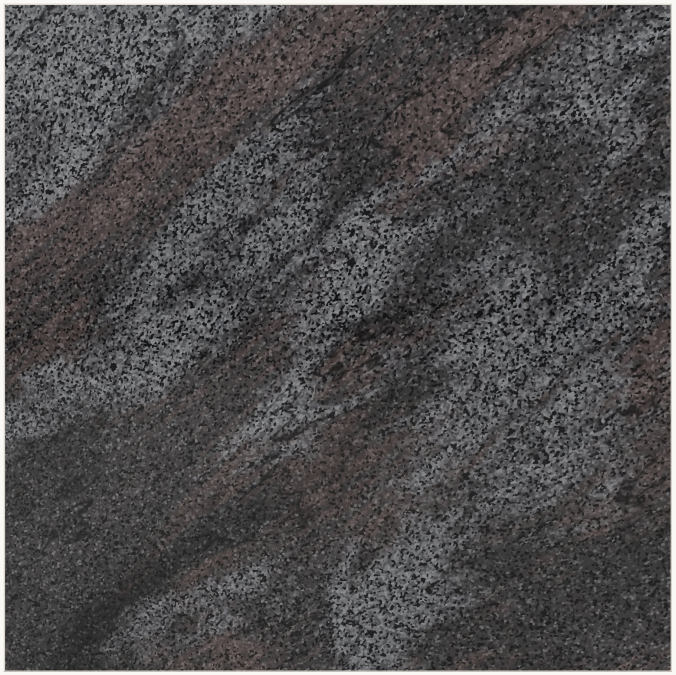
Salt & Pepper — surface detail

same physical patch, then lit close-up

PHOTO PATCH ~23 cm wide



SHADER ALBEDO 2048 px · same area



LIT CLOSE-UP 2400 px · polished face

Observed in the photographs

- Neutral charcoal base with the most grey "salt-and-pepper" clouds of the six.
- Dusty-rose layers are weak; black biotite clots are large and frequent.
- Flow rises ~32° to the right with gentle sweeps.

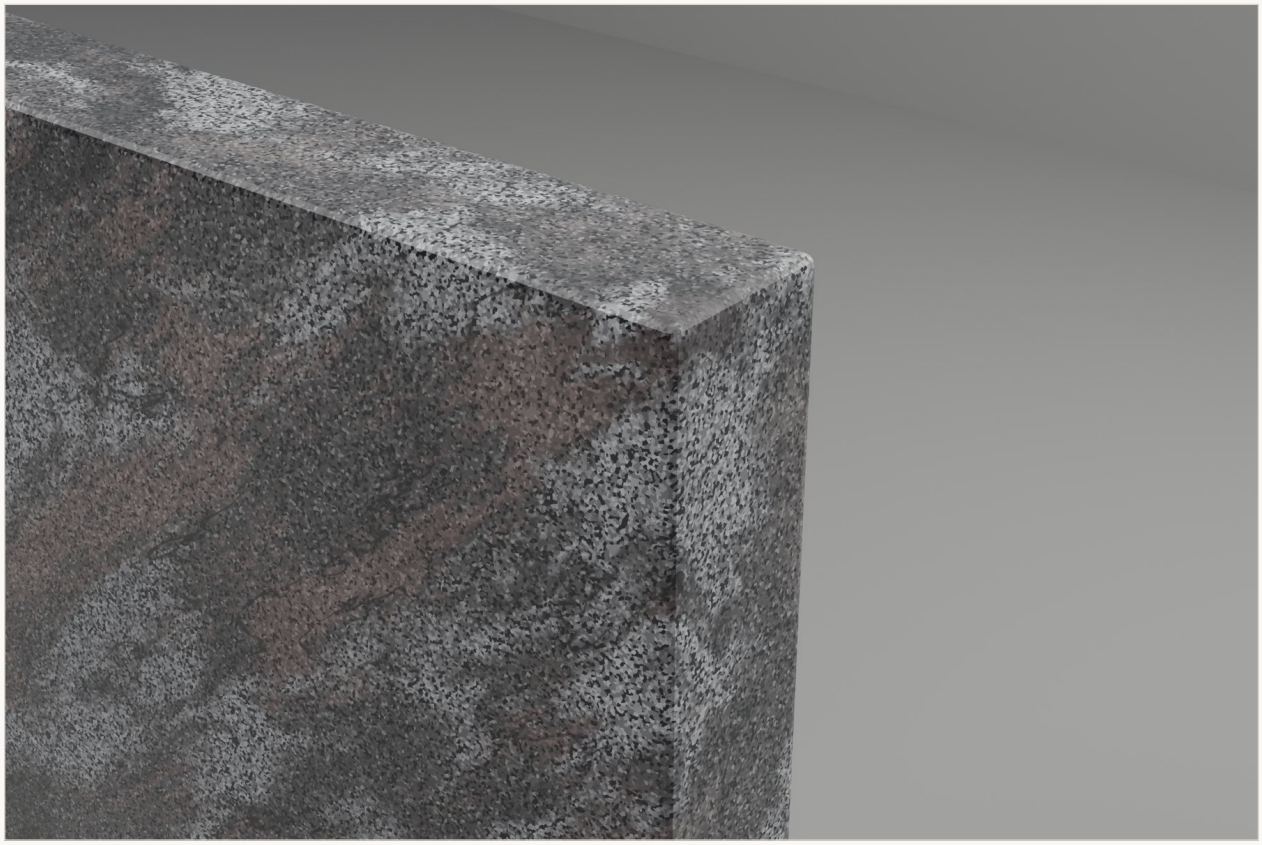
Calibration (photo → shader)

Mean sRGB	58 55 55	→	**58 54 55**
Contrast σ	27.6	→	**23.2**
p5–p95	10–101	→	**13–96**
Reddish pixels	10%	→	**9%**

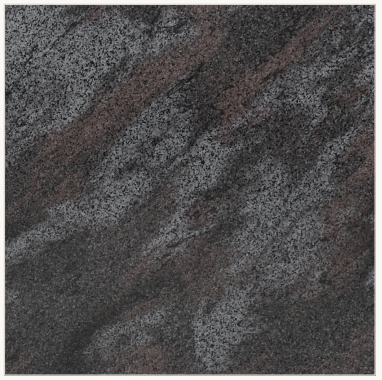
DATASET 01 · FOLDER 1st · REF 269567

Salt & Pepper — finish and seed variation

edge close-up · same material, four seeds



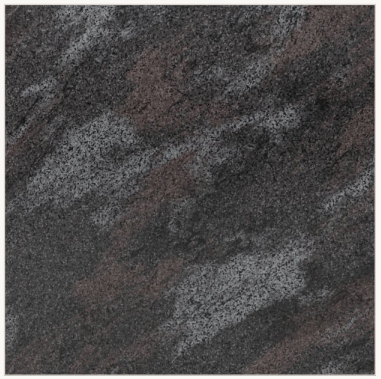
HONED EDGE polished face meets the side / top finish · 2400 × 1600



SEED 1 (DEFAULT)

SEED 11

SEED 222



SEED 3333

Finish

The wave-cut side panels are honed: the pattern stays visible but lifts to a light, cool grey.

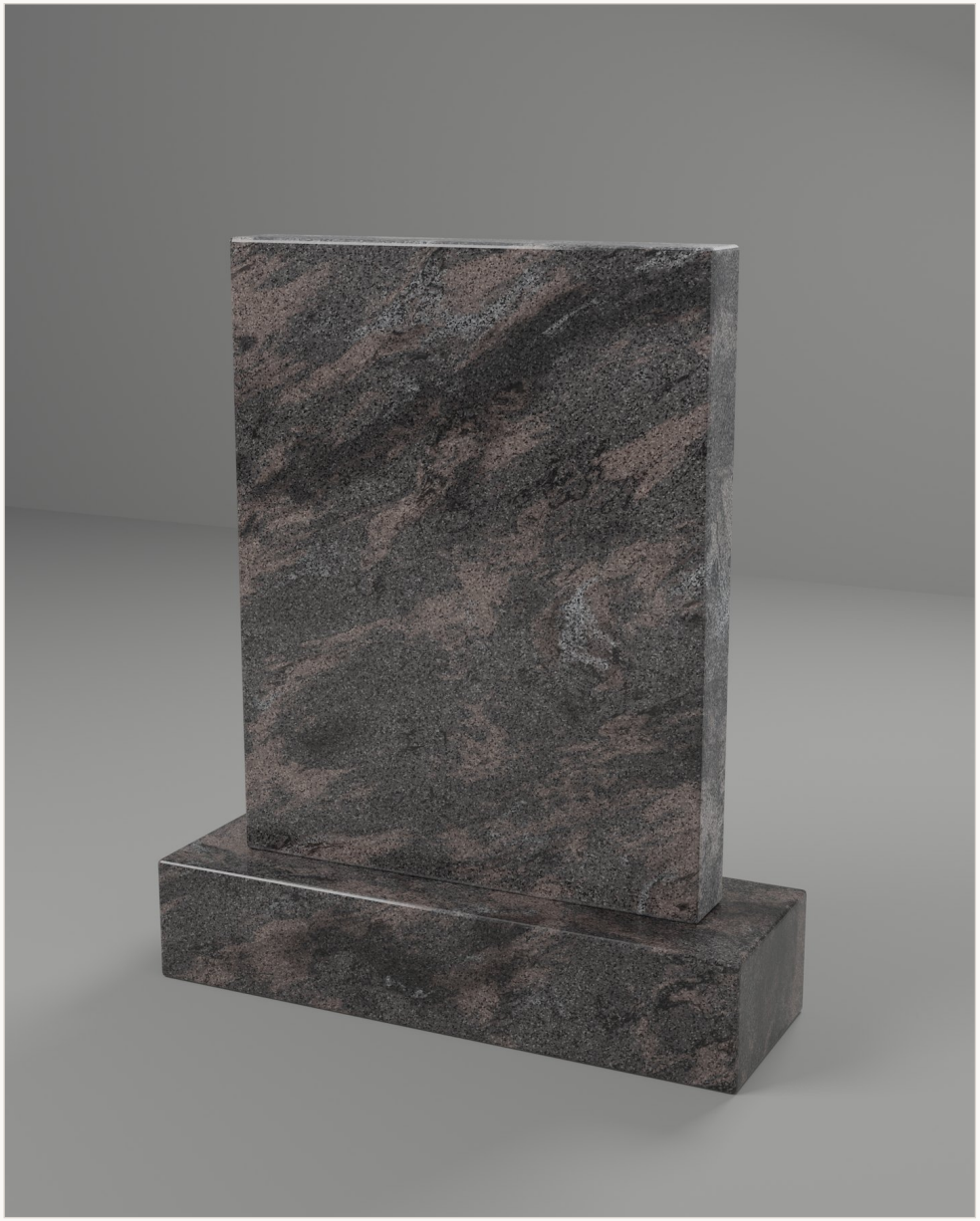
Key preset values

scale	**9**	flow angle	**32°**	stretch	**0.35**	warp	**0.9**
fold	**0.35**	rose	**0.38**	leucosome	**0.7**	melanosome	**0.45**
biotite	**0.6**	speckle	**0.6**	lamination	**0.1**	saturation	**0.7**

Flame Swirl

GS02_FlameSwirl · seed 2

REFERENCE PHOTOGRAPH folder 2nd · original 12 MP



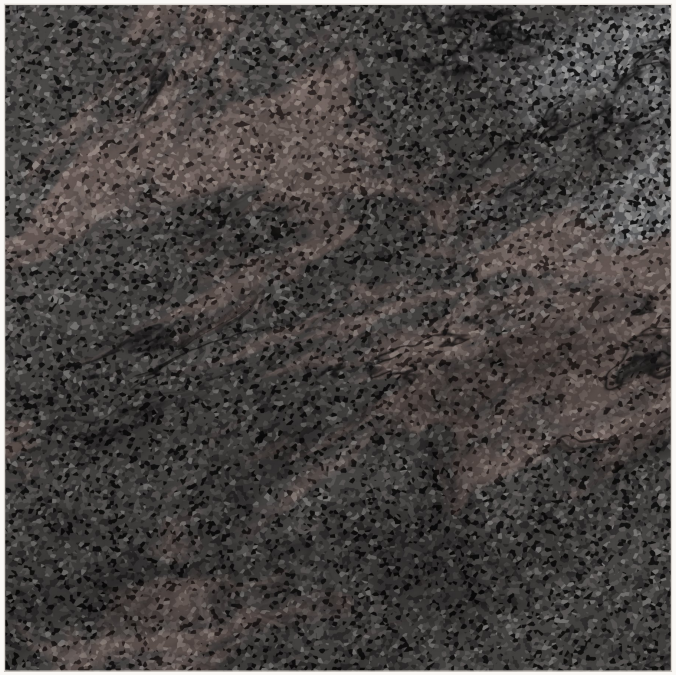
PROCEDURAL MATERIAL Cycles render · 2400 × 3000 · polished + honed

DATASET 02 · FOLDER 2nd · REF 269621

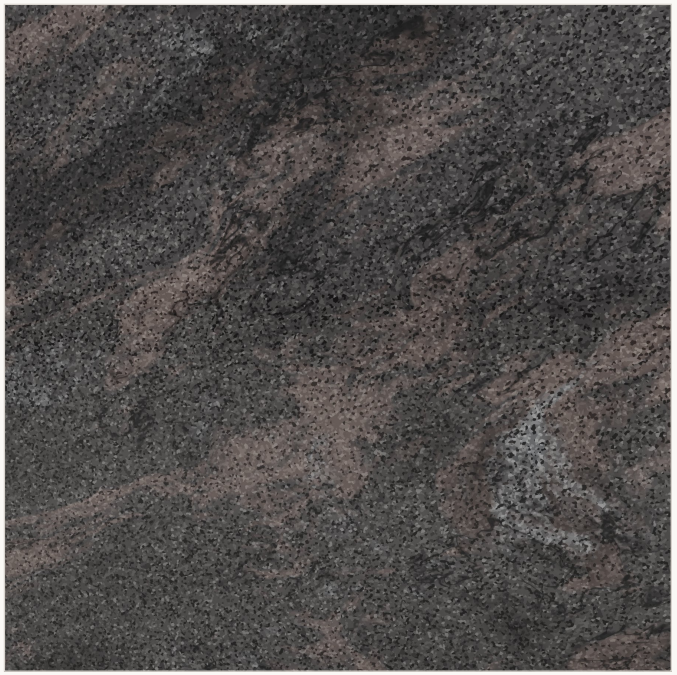
Flame Swirl — surface detail

same physical patch, then lit close-up

PHOTO PATCH ~27 cm wide



SHADER ALBEDO 2048 px · same area



LIT CLOSE-UP 2400 px · polished face

Observed in the photographs

- Warmer, brighter grey-brown stone with soft rose-brown clouds.
- Fine, even speckle; thin crinkly black schlieren lines.
- Shallow flow (~18°) with strong large-scale sweeps.

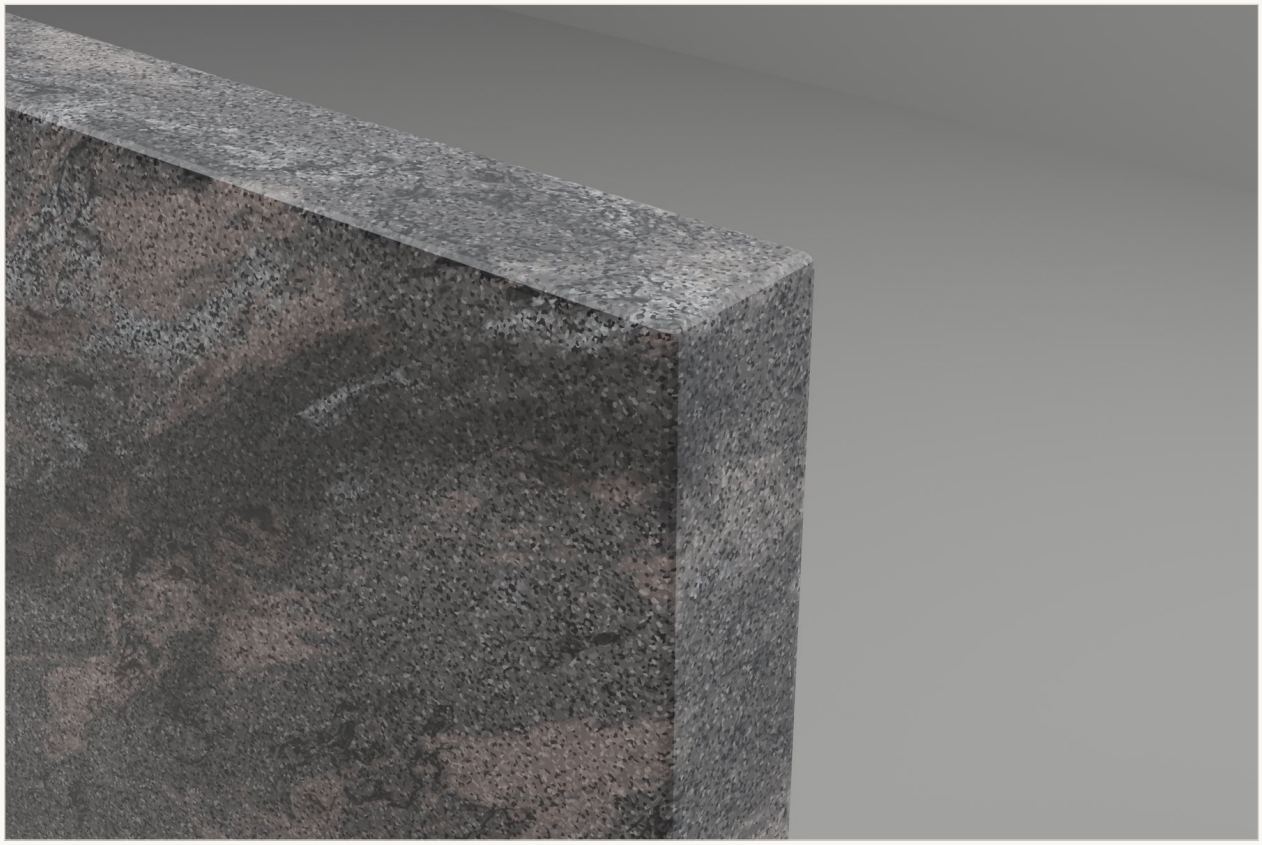
Calibration (photo → shader)

Mean sRGB	81 76 75	→	**76 71 70**
Contrast σ	22.5	→	**23.4**
p5–p95	41–114	→	**31–112**
Reddish pixels	18%	→	**24%**

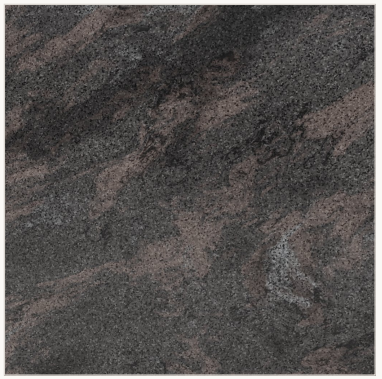
DATASET 02 · FOLDER 2nd · REF 269621

Flame Swirl — finish and seed variation

edge close-up · same material, four seeds

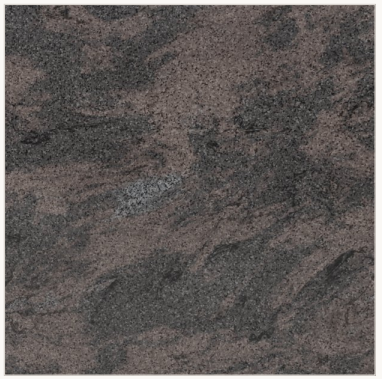


HONED EDGE polished face meets the side / top finish · 2400 × 1600

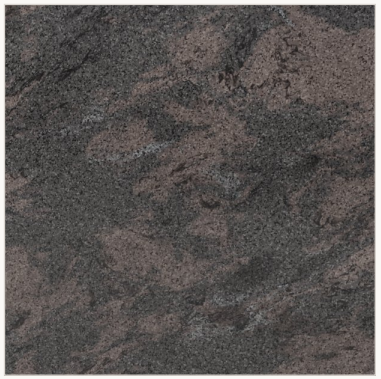


SEED 2 (DEFAULT)

SEED 11



SEED 222



SEED 3333

Finish

The flame cut-out and recessed ribbons are honed: cool blue-grey, matched to the photo at sRGB ~123/126/129.

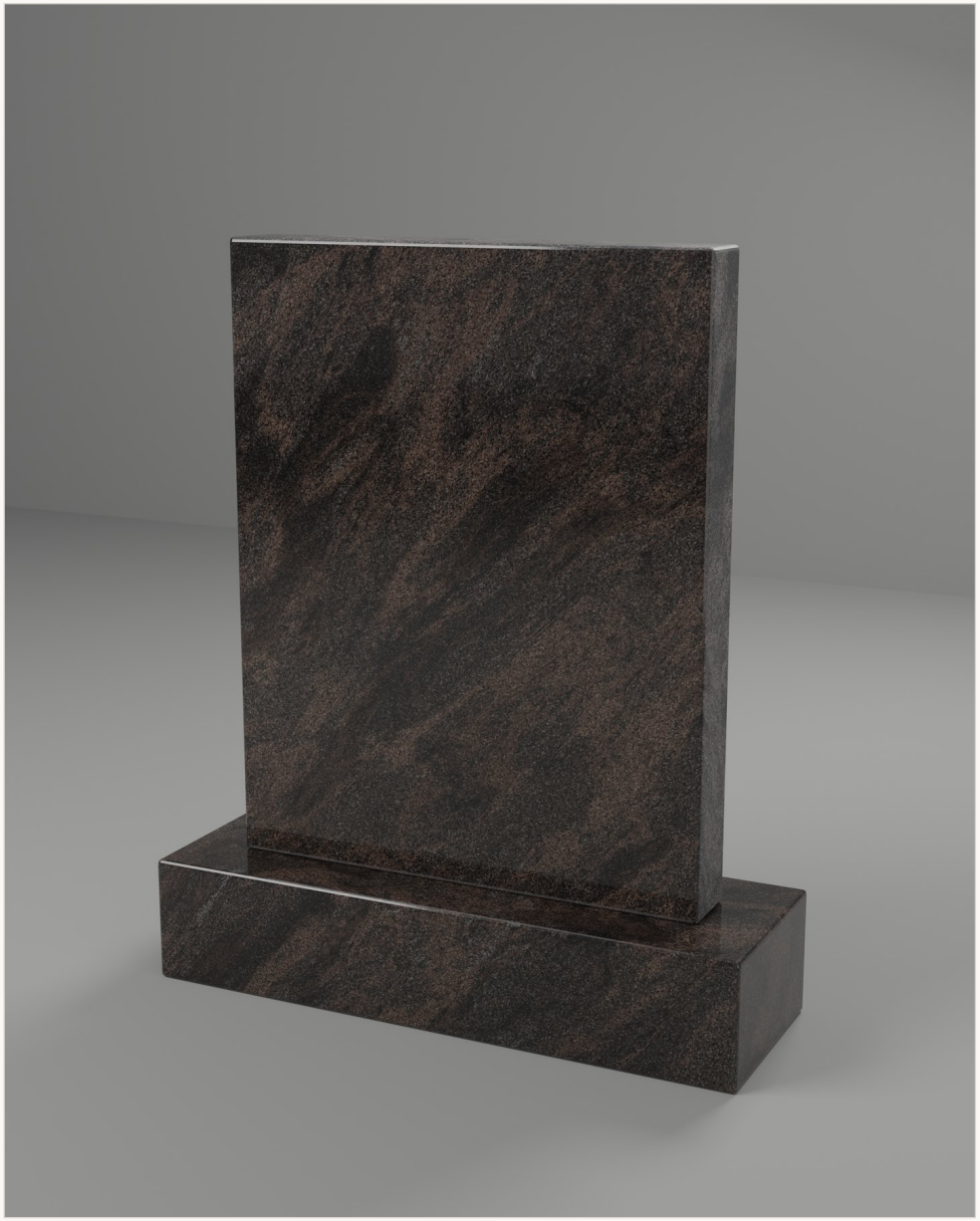
Key preset values

scale	**8**	flow angle	**18°**	stretch	**0.5**	warp	**2**
fold	**0.7**	rose	**0.45**	leucosome	**0.25**	melanosome	**0.3**
biotite	**0.9**	speckle	**0.3**	lamination	**0**	saturation	**0.6**

Dark Maroon

GS03_DarkMaroon · seed 3

REFERENCE PHOTOGRAPH folder 3rd · original 12 MP



PROCEDURAL MATERIAL Cycles render · 2400 × 3000 · polished + honed

DATASET 03 · FOLDER 3rd · REF SE269510

Dark Maroon — surface detail

same physical patch, then lit close-up

PHOTO PATCH ~31 cm wide

SHADER ALBEDO 2048 px · same area

LIT CLOSE-UP 2400 px · polished face

Observed in the photographs

- Darkest and warmest set: deep maroon-brown with almost no grey clouds.
- Strong folds with curling black bands and fine foliation stripes.
- Flow rises ~45°.

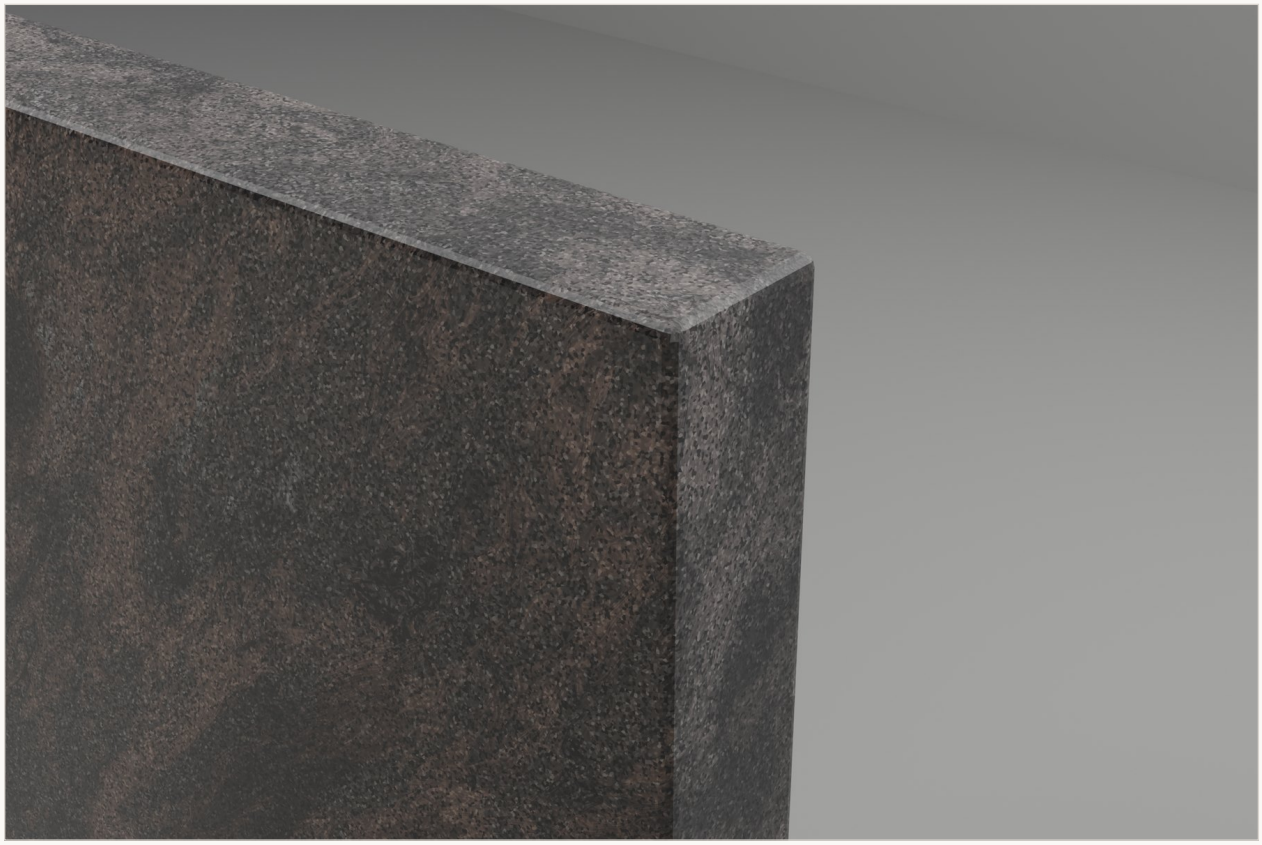
Calibration (photo → shader)

Mean sRGB	47 41 36	→	**49 41 37**
Contrast σ	18.3	→	**17.1**
p5–p95	10–71	→	**13–71**
Reddish pixels	52%	→	**51%**

DATASET 03 · FOLDER 3rd · REF SE269510

Dark Maroon — finish and seed variation

edge close-up · same material, four seeds



HONED EDGE polished face meets the side / top finish · 2400 × 1600

SEED 3 (DEFAULT)

SEED 11

SEED 222

SEED 3333

Finish

The wave side panel is honed (light grey); the face is deep, glossy maroon.

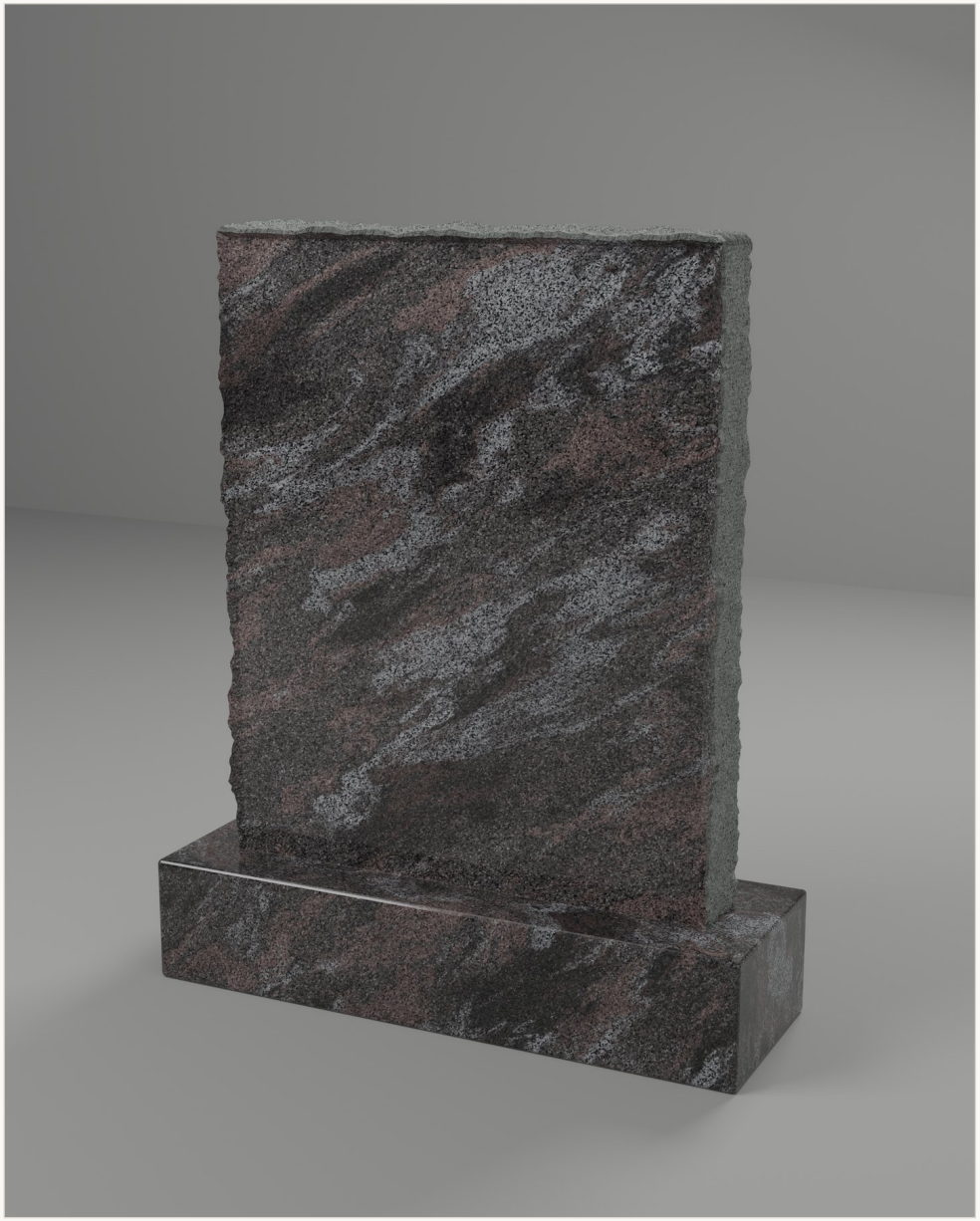
Key preset values

scale	**10**	flow angle	**45°**	stretch	**0.35**	warp	**2**
fold	**2.4**	rose	**0.55**	leucosome	**0.15**	melanosome	**0.55**
biotite	**0.85**	speckle	**0.2**	lamination	**0.6**	saturation	**0.85**

Folded + Rock-pitched

GS04_FoldedRock · seed 4

REFERENCE PHOTOGRAPH folder 4th · original 12 MP



PROCEDURAL MATERIAL Cycles render · 2400 × 3000 · polished + rock-pitched

DATASET 04 · FOLDER 4th · REF SE269539

Folded + Rock-pitched — surface detail

same physical patch, then lit close-up

PHOTO PATCH ~25 cm wide

SHADER ALBEDO 2048 px · same area

LIT CLOSE-UP 2400 px · polished face

Observed in the photographs

- Neutral charcoal with tight, swirling (ptygmatic) folds.
- Grey speckled clouds plus laminated purple-brown fold zones.
- Rock-pitched top edge (true displacement in Cycles).

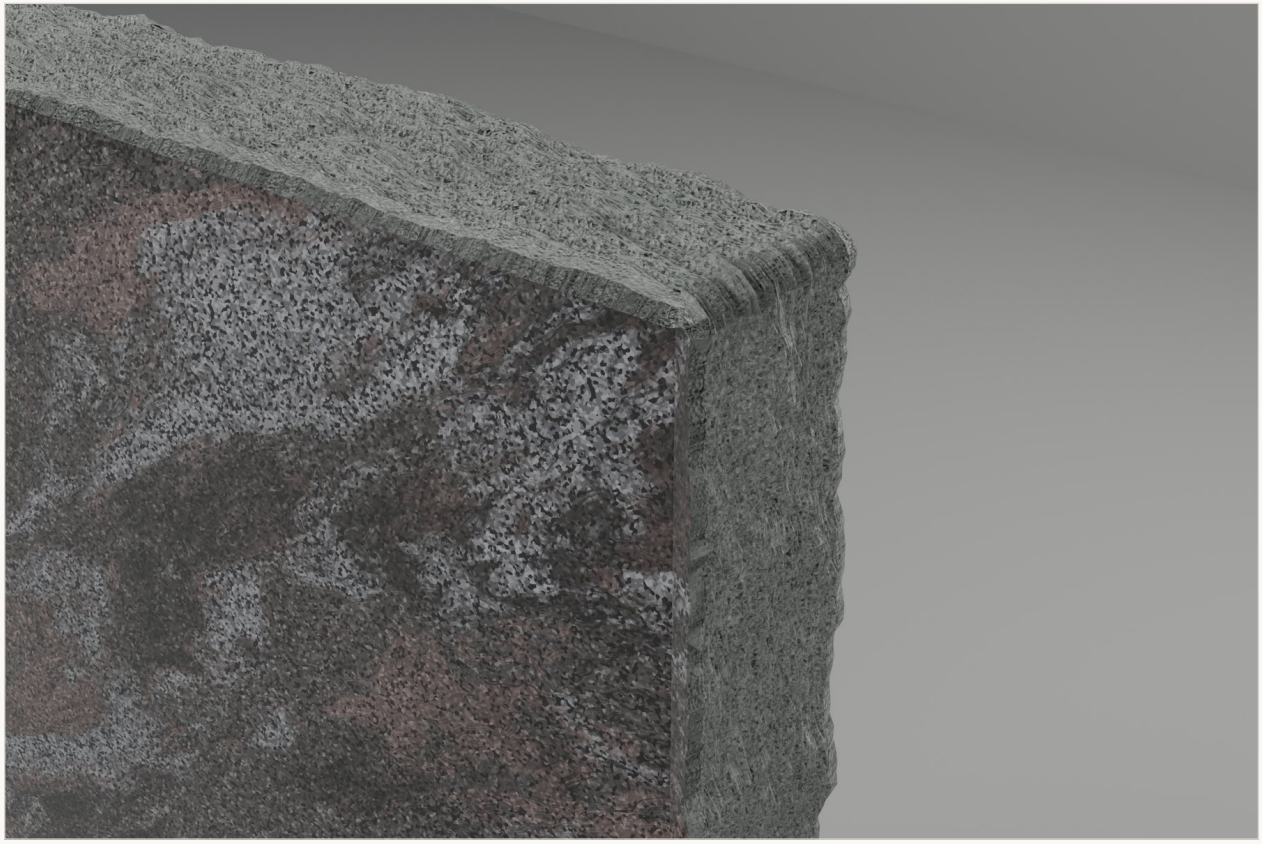
Calibration (photo → shader)

Mean sRGB	59 57 56	→	**55 52 51**
Contrast σ	25.7	→	**23.5**
p5–p95	13–98	→	**12–93**
Reddish pixels	9%	→	**12%**

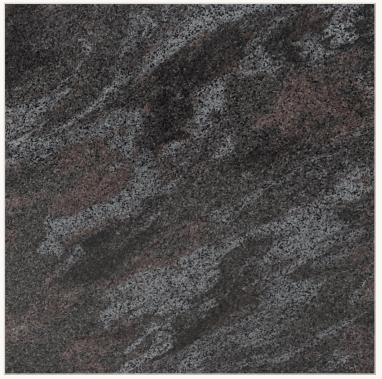
DATASET 04 · FOLDER 4th · REF SE269539

Folded + Rock-pitched — finish and seed variation

edge close-up · same material, four seeds

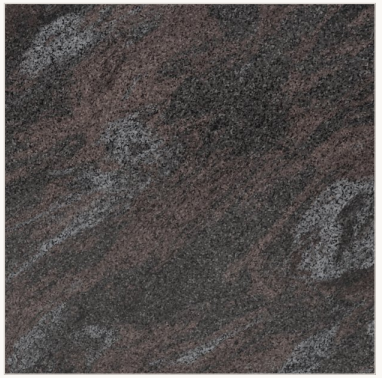


ROCK-PITCHED EDGE polished face meets the side / top finish · 2400 × 1600

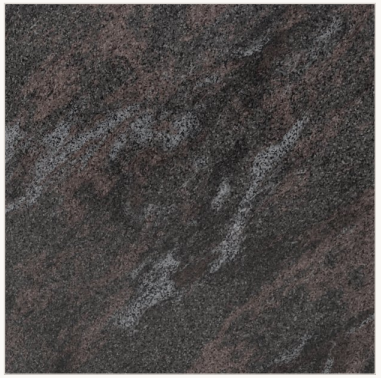


SEED 4 (DEFAULT)

SEED 11



SEED 222



SEED 3333

Finish

The top and side edges are rock-pitched: split, near-white crystals with 1–3 cm of real displaced relief.

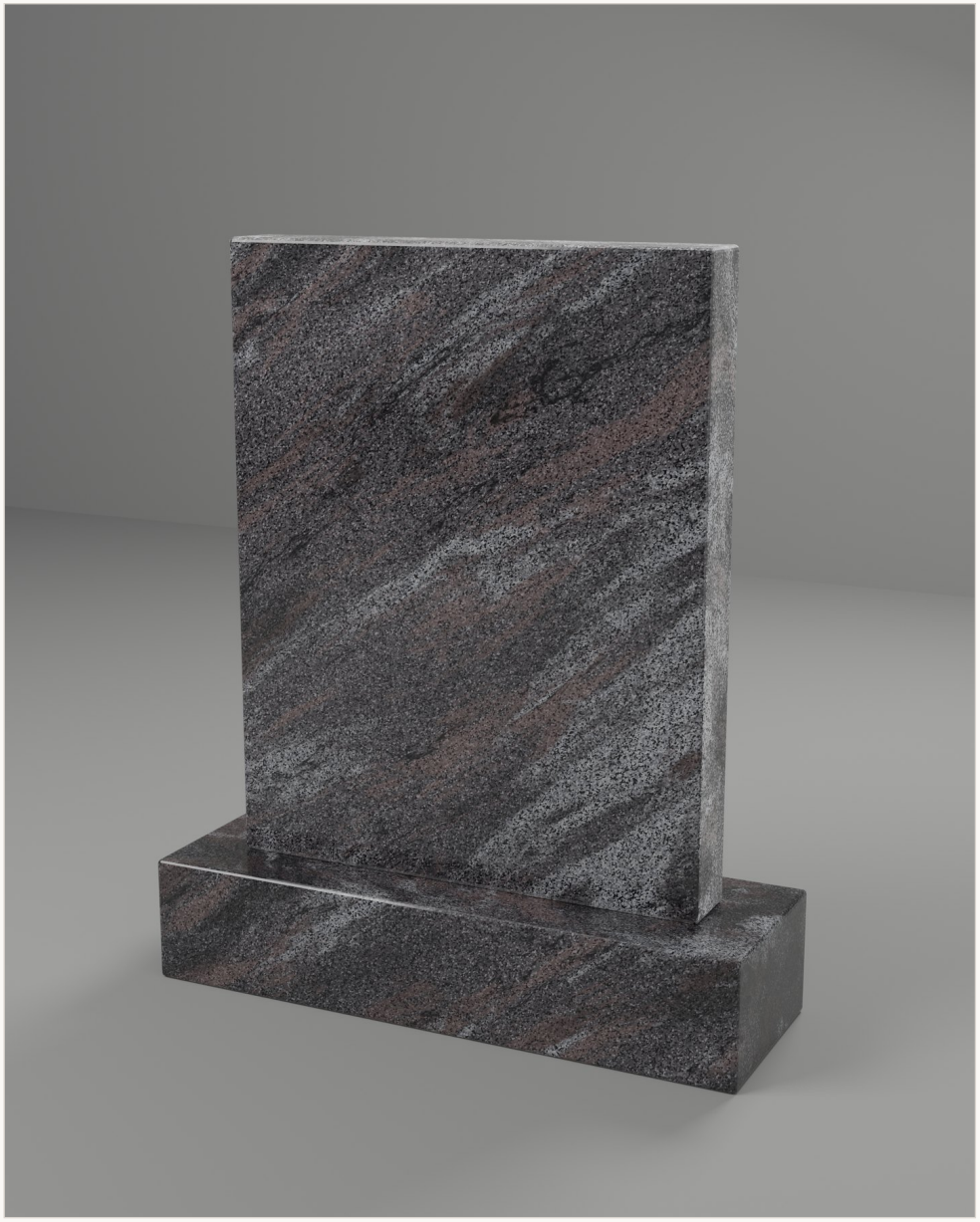
Key preset values

scale	**8.5**	flow angle	**33°**	stretch	**0.35**	warp	**1.8**
fold	**2.2**	rose	**0.42**	leucosome	**0.55**	melanosome	**0.5**
biotite	**0.6**	speckle	**0.5**	lamination	**0.7**	saturation	**0.85**

High Contrast

GS05_HighContrast · seed 5

REFERENCE PHOTOGRAPH folder 5th · original 12 MP



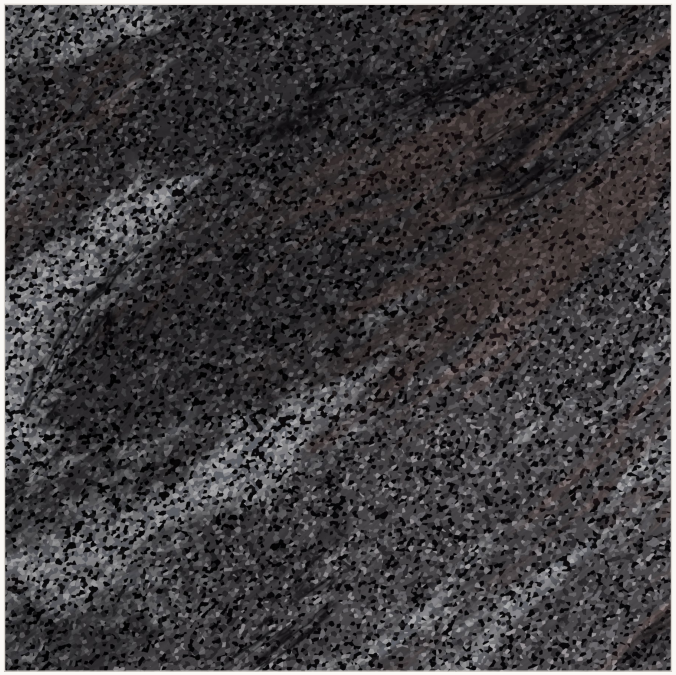
PROCEDURAL MATERIAL Cycles render · 2400 × 3000 · polished + honed

DATASET 05 · FOLDER 5th · REF SE269547

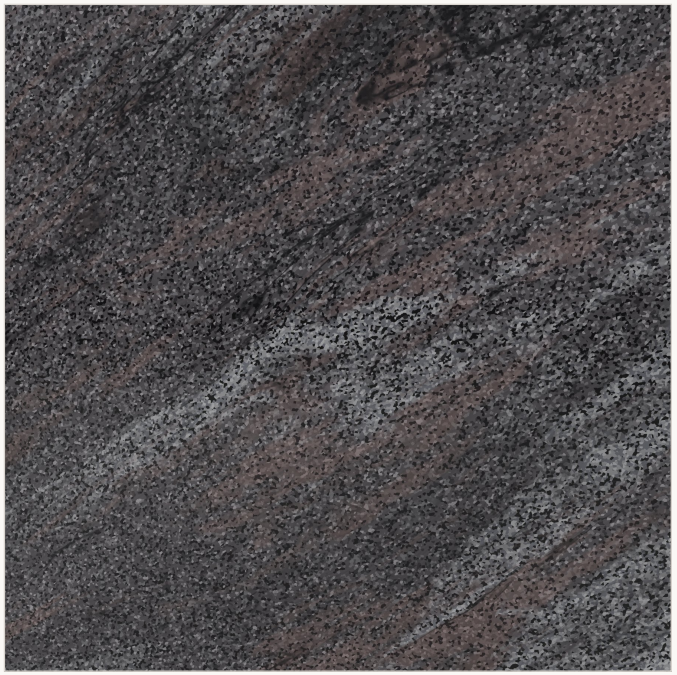
High Contrast — surface detail

same physical patch, then lit close-up

PHOTO PATCH ~29 cm wide



SHADER ALBEDO 2048 px · same area



LIT CLOSE-UP 2400 px · polished face

Observed in the photographs

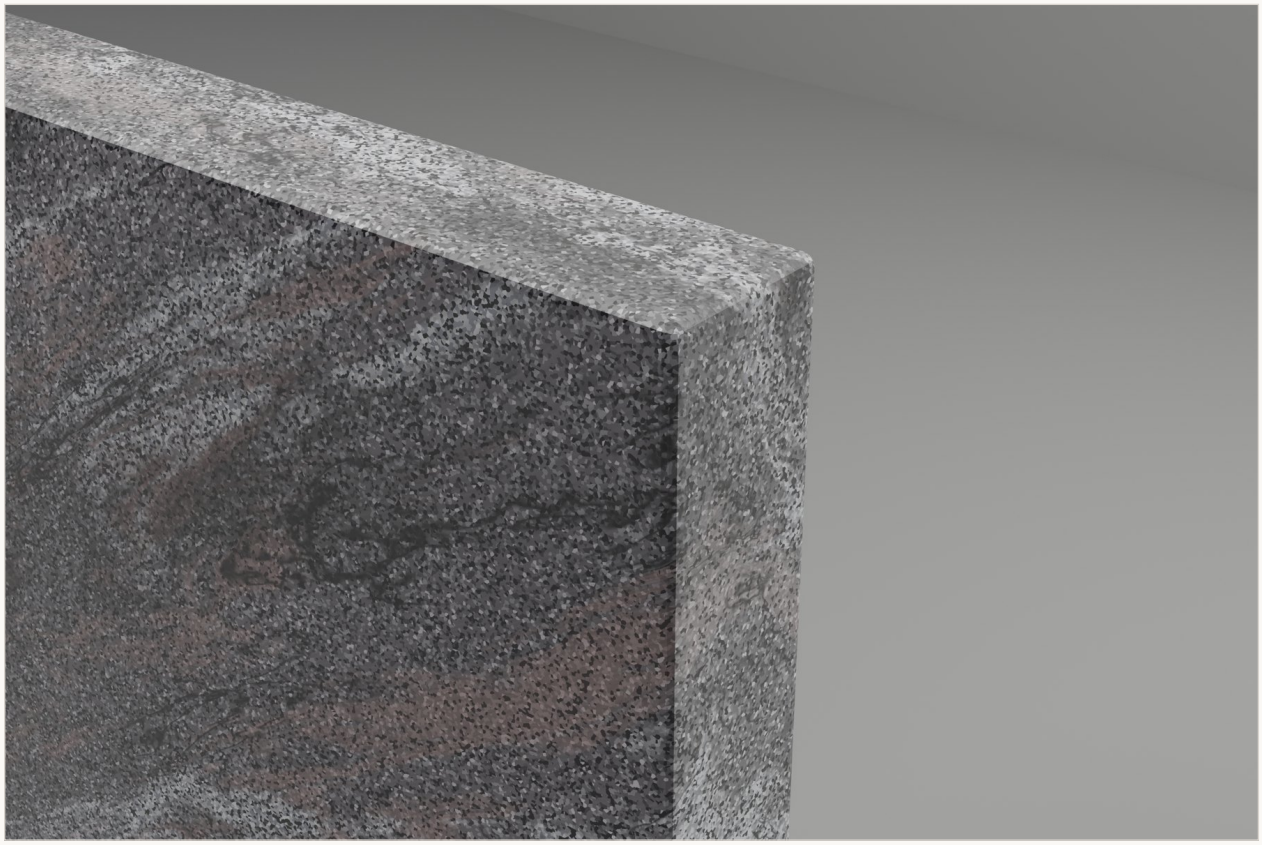
- Cool violet-charcoal with the strongest contrast of the six.
- Bright speckled clouds next to dense black biotite swirls.
- Flow ~25°, very swirly (low anisotropy).

Calibration (photo → shader)

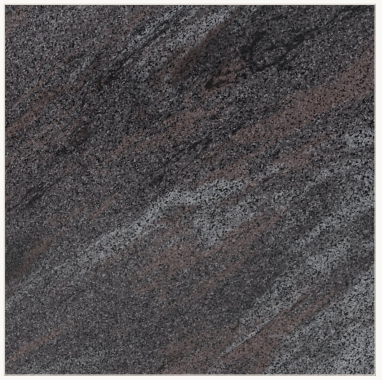
Mean sRGB	69 66 68	→	**73 69 72**
Contrast σ	33.9	→	**31.6**
p5–p95	11–125	→	**16–129**
Reddish pixels	15%	→	**16%**

High Contrast — finish and seed variation

edge close-up · same material, four seeds

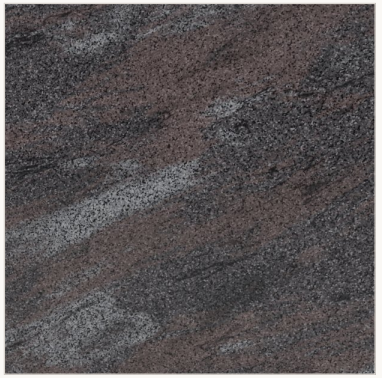


HONED EDGE polished face meets the side / top finish · 2400 × 1600

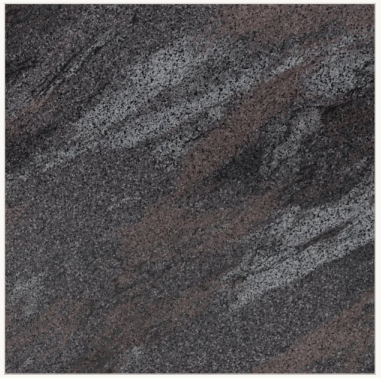


SEED 5 (DEFAULT)

SEED 11



SEED 222



SEED 3333

Finish

The flame ribbons are honed to a pale, warm grey (photo ~172/168/163, shader ~159/156/157).

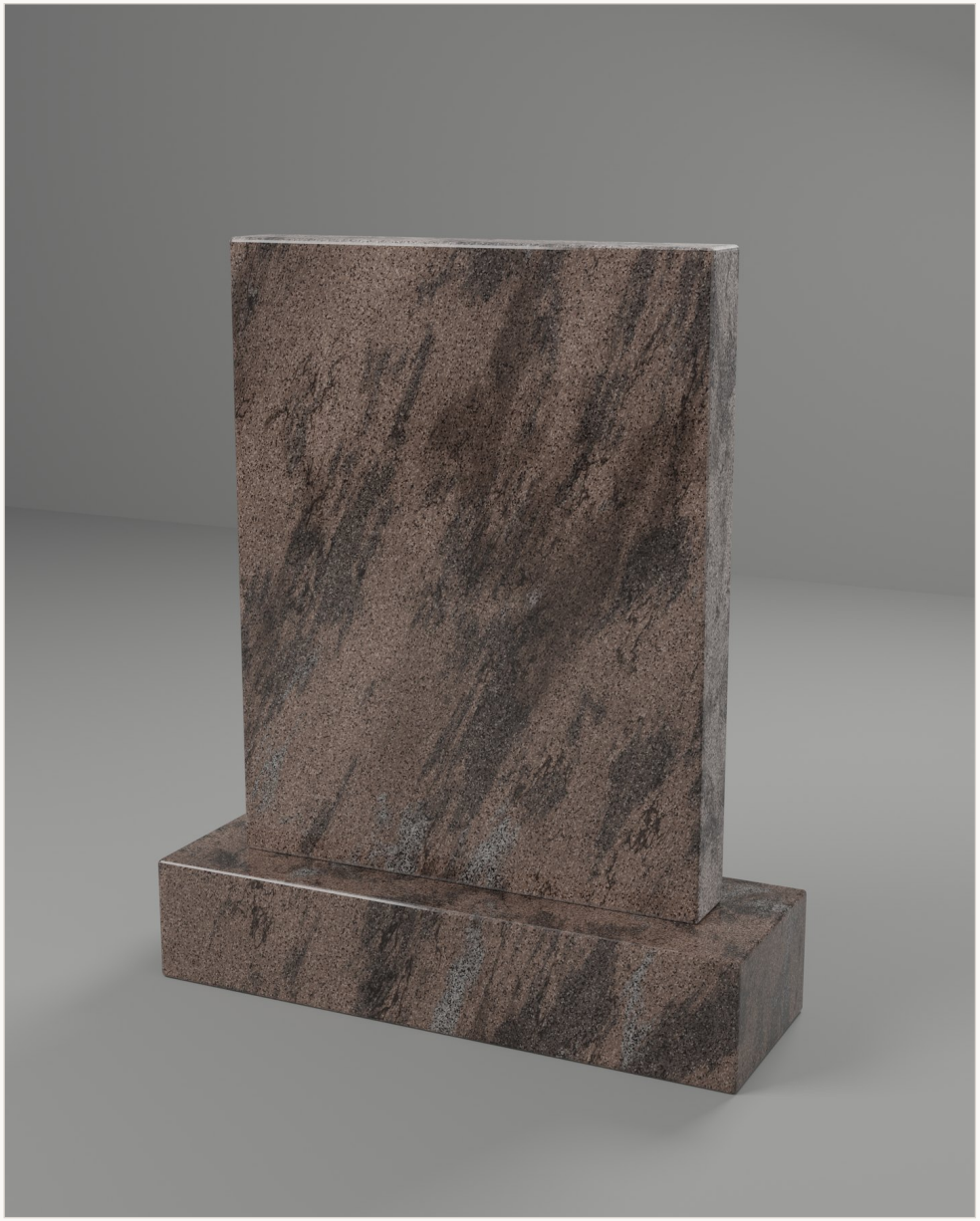
Key preset values

scale	**8.5**	flow angle	**25°**	stretch	**0.26**	warp	**1.3**
fold	**0.8**	rose	**0.42**	leucosome	**0.62**	melanosome	**0.6**
biotite	**0.75**	speckle	**0.7**	lamination	**0.25**	saturation	**1**

Rust Red

GS06_RustRed · seed 6

REFERENCE PHOTOGRAPH folder 6th · original 12 MP



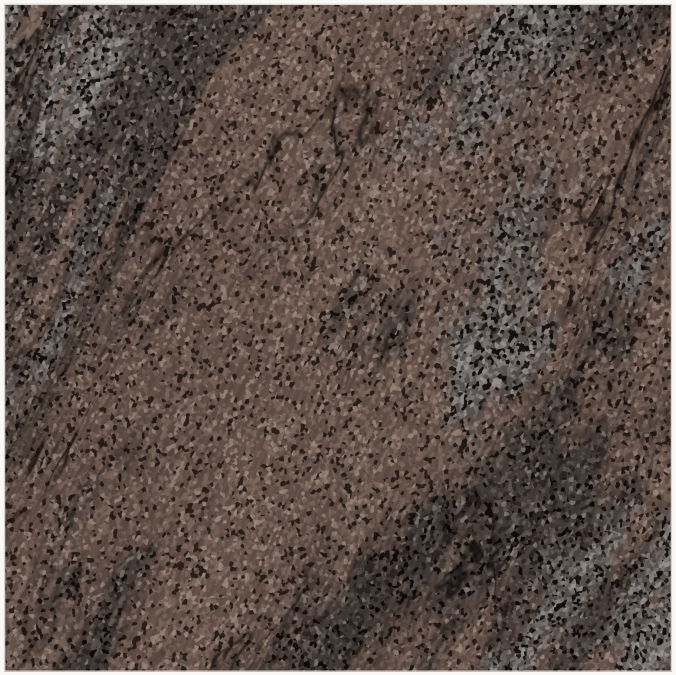
PROCEDURAL MATERIAL Cycles render · 2400 × 3000 · polished + honed

DATASET 06 · FOLDER 6th · REF SE269550

Rust Red — surface detail

same physical patch, then lit close-up

PHOTO PATCH ~28 cm wide



SHADER ALBEDO 2048 px · same area

LIT CLOSE-UP 2400 px · polished face

Observed in the photographs

- Rust and red-brown stone: 94% of pixels lean red.
- Almost no grey clouds; long, steep dark streaks (~65°).
- Fine foliation stripes inside the rust layers.

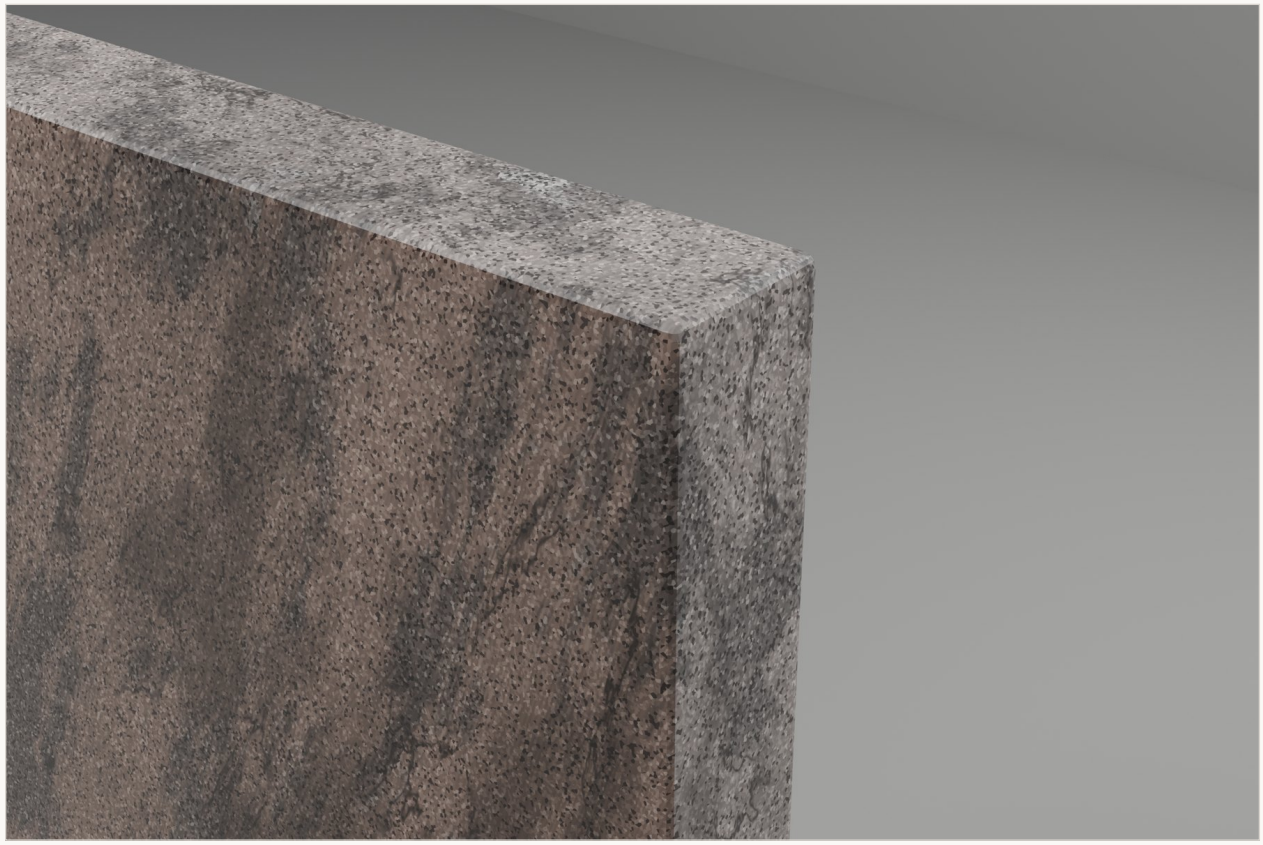
Calibration (photo → shader)

Mean sRGB	82 69 62	→	**84 71 65**
Contrast σ	27.1	→	**24.3**
p5–p95	27–116	→	**31–113**
Reddish pixels	94%	→	**93%**

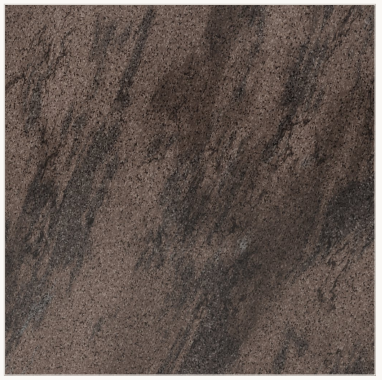
DATASET 06 · FOLDER 6th · REF SE269550

Rust Red — finish and seed variation

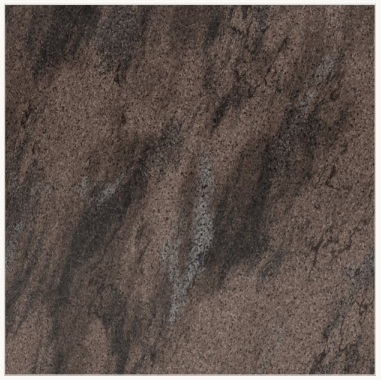
edge close-up · same material, four seeds



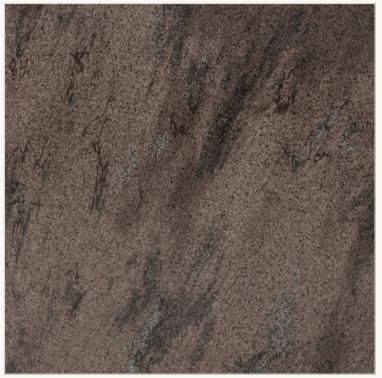
HONED EDGE polished face meets the side / top finish · 2400 × 1600



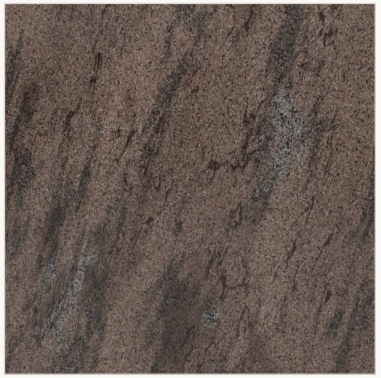
SEED 6 (DEFAULT)



SEED 11



SEED 222



SEED 3333

Finish

Fully polished in the photos; a honed variant is included for sawn or matte faces.

Key preset values

scale	**9**	flow angle	**65°**	stretch	**0.35**	warp	**1.2**
fold	**0.35**	rose	**0.8**	leucosome	**0.25**	melanosome	**0.4**
biotite	**0.75**	speckle	**0.2**	lamination	**0.35**	saturation	**0.75**

Seed control and parameters

one self-contained script per dataset folder

Run a script

- Open Blender 4.0+ (tested on 5.2 LTS) → Scripting → open the .py in the dataset folder → Run Script.
- Selected meshes get <ID>_Polished; the Honed (and, for 04, RockPitched) materials are created alongside.
- Object coordinates in metres, front face in the X/Z plane. Apply scale (Ctrl+A).

Seed control (three levels, they add up)

- Material: the "Granite" group node → Seed (live), SEED in the script, or set_seed(42).
- Per object: Custom Property granite_seed, set_object_seed(obj, 7) or randomize_selected_seeds().
- Automatic: Object Random Seed = 1 makes duplicates differ with no setup; 0 makes them identical.
- Seed Variation (0–1) also lets the seed nudge warp and the amount of rose / grey clouds.

Verified in Blender: same seed → correlation 0.996; different seed → 0.08; re-entering a seed reproduces the image exactly.

Preset values per dataset

	01	02	03	04	05	06
scale	9	8	10	8.5	8.5	9
flow °	32	18	45	33	25	65
stretch	0.35	0.5	0.35	0.35	0.26	0.35
warp	0.9	2	2	1.8	1.3	1.2
fold	0.35	0.7	2.4	2.2	0.8	0.35
crinkle	0.4	1.8	0.8	0.8	0.5	0.5
rose	0.38	0.45	0.55	0.42	0.42	0.8
leucosome	0.7	0.25	0.15	0.55	0.62	0.25
melanosome	0.45	0.3	0.55	0.5	0.6	0.4
biotite	0.6	0.9	0.85	0.6	0.75	0.75
speckle	0.6	0.3	0.2	0.5	0.7	0.2
lamination	0.1	0	0.6	0.7	0.25	0.35
grain	75	80	75	75	75	75
saturation	0.7	0.6	0.85	0.85	1	0.75
brightness	1	1.08	1	1.1	0.92	1
honed gain	1.2	1.2	1.2	1.2	3	1.2

Every value is also an input on the Granite node group, so the look can be tuned live without editing code.